

Nova Charging and Manufacturing Acceptance Report

Report scope

Requirements evaluated: 12

Domains: Charging, Manufacturing quality

Methods: bench test, environmental exposure, inspection, log review

Result vocabulary: PASS, FAIL, NOT TESTED, INCONCLUSIVE

1. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-CHG-001 AC input range	PASS - Executed for REQ-NVA-CHG-001; the recorded result confirms: Rated power is delivered at 180 V AC. NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-001: Rated power is delivered at 264 V AC. NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-001: Rated power is delivered at 45 Hz. NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-001: Rated power is delivered at 65 Hz.	Bench / inspection record Reviewed by DV team
REQ-NVA-CHG-002 Conversion efficiency	PASS - Executed for REQ-NVA-CHG-002; the recorded result confirms: Battery voltage is 400 V. PASS - Executed for REQ-NVA-CHG-002; the recorded result confirms: Charger output power is 11 kW. FAIL - Executed for REQ-NVA-CHG-002; the observed result does not satisfy: Charger efficiency is at least 94 percent.	Bench / inspection record Reviewed by DV team
REQ-NVA-CHG-003 Power factor	NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-003: Charger load is at least 50 percent. NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-003: Charger load does not exceed rated output. NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-003: Input power factor is at least 0.99.	Bench / inspection record Reviewed by DV team
REQ-NVA-CHG-004 Connector temperature	INCONCLUSIVE - Work was attempted for REQ-NVA-CHG-004, but incomplete scope or calibration prevents deciding: DC charging current is 200 A. INCONCLUSIVE - Work was attempted for REQ-NVA-CHG-004, but incomplete scope or calibration prevents deciding: Maximum connector terminal temperature is 90 deg C or less. INCONCLUSIVE - Work was attempted for REQ-NVA-CHG-004, but incomplete scope or calibration prevents deciding: Terminal temperature spread is 8 deg C or less. INCONCLUSIVE - Work was attempted for REQ-NVA-CHG-004, but incomplete scope or calibration prevents deciding: Charging continues for 20 minutes.	Bench / inspection record Reviewed by DV team

2. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-CHG-005 Isolation-fault response	PASS - Executed for REQ-NVA-CHG-005; the recorded result confirms: Charging isolation falls below 100 kohm. PASS - Executed for REQ-NVA-CHG-005; the recorded result confirms: Energy transfer stops within 100 ms. PASS - Executed for REQ-NVA-CHG-005; the recorded result confirms: Input relays open. PASS - Executed for REQ-NVA-CHG-005; the recorded result confirms: Diagnostic CHG-ISO-01 is reported.	Bench / inspection record Reviewed by DV team
REQ-NVA-QLT-001 Instrument calibration	PASS - Executed for REQ-NVA-QLT-001; the recorded result confirms: Every acceptance instrument is identified by serial number. FAIL - Executed for REQ-NVA-QLT-001; the observed result does not satisfy: Calibration expiry is later than the test date.	Bench / inspection record Reviewed by DV team
REQ-NVA-QLT-002 Software traceability	NOT TESTED - No test or inspection was performed for REQ-NVA-QLT-002: Released controller records a software part number. NOT TESTED - No test or inspection was performed for REQ-NVA-QLT-002: Recorded build hash matches the approved manifest.	Bench / inspection record Reviewed by DV team
REQ-NVA-QLT-003 Sampling records	INCONCLUSIVE - Work was attempted for REQ-NVA-QLT-003, but incomplete scope or calibration prevents deciding: Inspection record identifies the production lot. INCONCLUSIVE - Work was attempted for REQ-NVA-QLT-003, but incomplete scope or calibration prevents deciding: Inspection record states the sample size. INCONCLUSIVE - Work was attempted for REQ-NVA-QLT-003, but incomplete scope or calibration prevents deciding: Inspection record names the sampling plan.	Bench / inspection record Reviewed by DV team

3. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-QLT-004 Weld acceptance	PASS - Executed for REQ-NVA-QLT-004; the recorded result confirms: Every inspected busbar weld is evaluated. PASS - Executed for REQ-NVA-QLT-004; the recorded result confirms: Effective weld diameter is at least 4.5 mm. PASS - Executed for REQ-NVA-QLT-004; the recorded result confirms: Weld porosity is below 8 percent. PASS - Executed for REQ-NVA-QLT-004; the recorded result confirms: No crack is visible at 10x magnification.	Bench / inspection record Reviewed by DV team
REQ-NVA-QLT-005 Label content	PASS - Executed for REQ-NVA-QLT-005; the recorded result confirms: Product label includes hardware revision. NOT TESTED - No test or inspection was performed for REQ-NVA-QLT-005: Product label includes software revision. NOT TESTED - No test or inspection was performed for REQ-NVA-QLT-005: Product label includes a machine-readable serial number. NOT TESTED - No test or inspection was performed for REQ-NVA-QLT-005: Machine-readable and human-readable serial numbers match.	Bench / inspection record Reviewed by DV team

4. Image-based inspection result

Visual Inspection Record

REQ-NVA-CHG-006 | Interlock sequence

PASS
the recorded result confirms: Proximity switch closes before pilot energization.

UNTESTED
NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-006: Pilot energizes before power contactors close.

UNTESTED
NOT TESTED - No test or inspection was performed for REQ-NVA-CHG-006: Every completed step is displayed.

REQ-NVA-QLT-006 | Sealant inspection

PASS
the recorded result confirms: Blue sealant bead forms a continuous closed loop.

PASS
the recorded result confirms: Sealant remains between both red tolerance boundaries.

FAILED
the observed result does not satisfy: No sealant gap exceeds 0.5 mm.

Image record NOVA-DV-VIS | Reviewer: QA-17 | Controlled copy

Figure 4-1. Recorded inspections for REQ-NVA-CHG-006, REQ-NVA-QLT-006; interpretation requires the raster content.